

NCIE-013 Skeletal v0.1 → Full v1.0

Expansion Verification Report

Functional Modules Specification — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Title Reconciliation Verification (Directive Item 3) — SOURCE-VERIFIED

Authoritative register used: NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), located on the controlled document store and confirmed as the applicable register because it explicitly supersedes the earlier Production-ready register (17 August 2026) and is the register cited by the NCIE-013 Skeletal v0.1's own framework-authority statement. NCIE-013 entry located: Section 3, 'Reconciled Master Documentation Register' table, row NCIE-013. Authoritative formal title (as recorded in the register): 'Functional Modules Specification'. Current specification title prior to this verification pass: 'NCIE-013 — Domain Functional Modules Specification'. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register's Primary Purpose text — 'Detailed specifications for Network, Traffic, Revenue, Mobile Money, SIM Registration, Anti-Fraud, REWS and related operational modules' — matches this document's actual Chapters 11–18 domain-module scope exactly; the only difference is the register's omission of the word 'Domain'; no material discrepancy in scope, responsibility or substance was found). Correction applied: the document-control title was corrected to 'NCIE-013 — Functional Modules Specification' in the cover, §0.1, the Document Governance table, the Authority Chain, the running header, §34.6 and core document properties, under a pre-approval surgical correction (not a v1.0→v1.1 amendment); no chapter content or domain-module architecture was altered. 34-chapter integrity after correction: PASS (34/34, re-verified). Salient Things to Note after correction: PASS (34/34, re-verified). Human Review Register reconciliation after correction: PASS (34 total / 30 Blocking / 4 Non-Blocking / 0 duplicate / 0 orphan — unchanged from the pre-correction baseline). Three-artifact consistency after correction: PASS. This finding is SOURCE-VERIFIED — established directly from the text of the Master Controlled Register v2.0 document itself, not inferred from the document's own naming convention, from the Skeletal v0.1's wording, or from conversational context. Page/section citations above are limited to the register's own internal section numbering (Section 3), since the source file carries no stable page numbering independent of rendering.

2. Scope Verification

NCIE-013 v1.0 is a controlled full expansion of the approved NCIE-013 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into cross-domain functional doctrine, module lifecycle, rule/KPI/anomaly/alert frameworks and the eight domain intelligence modules, without inventing regulatory rules, breach criteria, KPI catalogues, thresholds, materiality limits, fraud typologies, REWS indicators, score cutoffs, source precedence or enforcement actions absent source authority, and without fabricating data availability, source quality, operator feeds, field semantics, coverage or production readiness.

3. 34-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 34 sequential 'NCIE-013 — Chapter N Expansion' labels, N = 1..34, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure (the formal-title reconciliation in §1 above did not require any chapter renaming).

4. Salient Things to Note Hard-Gate Verification

Per directive item 71 — continuing the discipline established after the NCIE-009 Chapter-27 omission — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 34 chapters (N = 1 through 34), with zero missing.

5. Skeleton Fidelity Verification

Each of the 34 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

6. Threshold/Rule Fabrication Audit

A full-text scan (directive item 62) for invented KPI, risk, fraud, REWS or materiality thresholds, breach criteria, rule catalogues or automated sanctions found none — Chapter 6 §6.1-6.2 explicitly defers the KPI catalogue and thresholds to Human Review (HR13-6-1), Chapter 11 defers regulatory thresholds (HR13-11-1), Chapter 13 defers materiality (HR13-13-1), and Chapter 16 defers the fraud typology catalogue (HR13-16-1) — every concrete number or catalogue is a named Human Review item, never an invented default.

7. Governing Functional-Module Invariants Verification

The 18 invariants proposed in the approved Skeletal v0.1 ('Proposed Functional-Module Invariants') are restated unchanged in front matter §0.4, and are additionally cross-referenced — not restated — against the 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants, the 19 NCIE-010 Evidence/provenance/audit invariants, the 20 NCIE-011 integration invariants, the 19 NCIE-012 UX/UI invariants, and other upstream invariants already binding from NCIE-005/006/007. A full-text scan confirms every invariant is both stated once in §0.4 and applied by name in at least one operative chapter (e.g., ANALYTICAL SIGNAL ≠ EVIDENCE in Ch.21; ALERT ≠ FINDING in Ch.8; ANOMALY ≠ FRAUD in Ch.7, 16; MISSING DATA ≠ ZERO VALUE in Ch.29).

8. Domain/Upstream Boundary Verification

Chapter 1 Table 1 fixes what NCIE-013 does not own (canonical data, ARGUS behavior, Memory/Context semantics, Agent/VPF semantics, institutional governance, security enforcement, Evidence/provenance meaning, API contracts, UX semantics — all NCIE-001–012's). Every consequential domain state in Chapters 2–32 is expressed as an analytical state (per Chapter 2's Domain-State Matrix) with the accountable holder marked 'to be assigned,' never as NCIE-013 inventing regulatory Findings, institutional Decisions or policy authority of its own (directive item 8).

9. Finding/Decision Audit

A full-text scan (directive item 63) for wording that would convert an Alert, Situation, anomaly, score, AI/VPF or workflow state into a Finding, Decision or approval found none — Chapter 2 Table 3 (Domain-State Matrix) explicitly marks Finding and Decision as external NCIE-008 Ch.18 acts, never autonomously created by a domain module, and Chapter 9, 16, 23 and 24 each independently restate the relevant non-equivalence for case handoff, fraud referral, AI/Agent output and workflow state respectively.

10. Evidence Audit

Verified (directive item 64) that analytical signals, connector data, model/Agent outputs and dashboards are not automatically Evidence and that the NCIE-010 lifecycle is preserved: Chapter 21 Table 26 (Domain Evidence Matrix) fixes an explicit non-Evidence state for every analytical output class, and Chapter 2 Table 3 confirms Evidence status flows only through the NCIE-010 lifecycle, never through domain-signal generation alone.

11. Cross-Domain Privacy Audit

Verified (directive item 65) that correlation/fusion does not bypass NCIE-009 purpose, authorization, classification, DLP or protected-identity controls: Chapter 10 Table 14 (Cross-Domain Correlation Matrix) fixes an explicit prohibited-disclosure-expansion rule for every correlation class, confirming correlation never grants access to a source domain the requester was not already authorized to see.

12. Data-Quality Audit

Verified (directive item 66) that missing, late, conflicting or revised data cannot silently become zero, current or clean state and that uncertainty remains visible: Chapter 29 confirms MISSING DATA ≠ ZERO VALUE and LATE DATA ≠ CURRENT TRUTH, requiring an explicit qualifier on every affected computation rather than a silent default.

13. Domain-State, Module Catalogue and Rule/KPI Governance Matrix Verification

Chapter 2 Table 3 (Domain-State Matrix, directive item 54) distinguishes 9 states from observation through Decision, each with an authority/provenance caution and owning chapter. Chapter 3 Table 5 (Module Catalogue Matrix, directive item 55) maps 10 domain intelligence chapters (Network through Temporal/Forecast) to inputs, outputs, rules/KPIs, Alerts, Evidence interface, ARGUS/Agent use, APIs/connectors, UX and security basis. Chapter 6 Table 9 (Rule/KPI Governance Matrix, directive item 56) fixes the identity/versioning/provenance structure for rules, KPIs and baselines without inventing a single concrete threshold.

14. Cross-Domain Correlation and Domain Evidence Matrix Verification

Chapter 10 Table 14 (Cross-Domain Correlation Matrix, directive item 57) maps 3 correlation classes to permitted inputs, purpose basis, authorization boundary and prohibited disclosure expansion. Chapter 21 Table 26 (Domain Evidence Matrix, directive item 58) maps 5 analytical output classes to their Evidence relationship, provenance reference, gap/contradiction handling and an explicit non-Evidence state for each.

15. Test-Status Integrity Verification

Chapter 32 Table 38 specifies 8 test classes (rule, KPI, anomaly, alert, cross-domain, Evidence traceability, security, degraded-data scenario), each traced to the chapter(s) it verifies. Every test is marked TEST SPECIFIED only; a full-text scan confirms no test anywhere in this document is marked TEST EXECUTED, TEST PASSED, SCENARIO VALIDATED, or claims IMPLEMENTATION EVIDENCE AVAILABLE (directive item 61) — detailed test-case authoring and execution are explicitly deferred to NCIE-016, and integration success (an NCIE-011 concern) is explicitly distinguished from domain-interpretation validity.

16. No Implementation Claims

A full-text scan confirms no module, rule, connector, model or threshold anywhere in this document is claimed implemented, calibrated, validated or production-ready merely because this v1.0 specifies it (directive item 67) — every RACI/control-ownership table marks its holder 'to be assigned,' and Chapter 32 explicitly marks every test SPECIFIED, not executed or scenario-validated.

17. Human Review Register Mechanical Reconciliation (Correct From First Build)

Chapter 33's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-32, plus HR13-33-1 (Chapter 33's own item) and HR13-34-1 (Chapter 34's own item) — both defined once and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 34 total items, 30 Blocking / 4 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 22 Institutional Confirmation Required, 5 Security/Sovereignty Confirmation Required (Chapters 10, 14, 15, 19, 27), 3 Legal/Policy Confirmation Required (Chapters 11, 13, 16), 3 Engineering/Infrastructure Discovery Required (Chapters 25, 29, 31), 1 Resolved by Upstream Baseline (Chapter 34).

18. Upstream Human Review Deduplication (Directive Item 60)

HR13-25-1 (standard versus bespoke domain connectors) restates the identical institutional question NCIE-011 already opened as HR11-16-1 (standard versus bespoke domain connectors, NCIE-011 Ch.16); Chapter 33 Table 41 cross-references HR11-16-1 rather than treating HR13-25-1 as an independent open question. Every other NCIE-013 Human Review item was individually checked against the NCIE-008–012 Human Review Registers and found to concern a domain-business-specific decision that those documents do not themselves resolve — no other duplicate was found.

19. Blocking Scope Discipline Verification

Each Blocking item in the Chapter 33 register states precisely which rule catalogue, threshold, typology, ownership assignment or acceptance authority it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 3, 25, 26, and the resolved Chapter 34 item) are explicitly marked as not gating the corresponding chapter's domain operation in the interim.

20. Traceability and Downstream Non-Preemption Verification

Chapter 34 Table 42 traces all 34 chapters to their primary upstream authority (approved NCIE-001-012 and/or the approved NCIE-013 Skeletal v0.1); no chapter states a domain-business requirement without an identifiable source. Table 43 confirms this document hands over operational responsibilities to NCIE-015 and the 8 domain test classes to NCIE-016 — without specifying their technical implementation, consistent with directive item 19 (downstream non-preemption).

21. Version, Document-Control and Running-Header Verification

Cover and Document Governance table confirmed reading: Edition 'In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)'; all ten upstream authority rows (NCIE-001 through NCIE-012) read their approved current versions; a 'Prior baseline' row records the approved Skeletal v0.1 as historically preserved; Status reads IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The running header — rendered from a copied NCIE-012 v1.0 template to inherit this series' visual styling, including the logo, centered cover text, page break before Document Governance, and non-splitting governance-table rows already corrected in that series — was corrected to 'NCIE-013 | DOMAIN FUNCTIONAL MODULES SPECIFICATION | VERSION 1.0' and the footer corrected from 'CONTROLLED UX DOCUMENT' to 'CONTROLLED DOMAIN DOCUMENT' and from 'NCIE-012' to 'NCIE-013' before this document was rendered to PDF.

22. Full-Document Cross-Reference Scan

Every occurrence of 'NCIE-012' in the final document was individually inspected (18 occurrences outside Upstream/Trace blocks). Each was confirmed to be a legitimate UX/workspace citation (e.g., 'NCIE-012 component library and state matrices', 'NCIE-012 workspace owner') consistent with directive item 8's requirement that NCIE-013 consume, not redefine, NCIE-012 UX semantics — none is a leftover donor-template branding error. The cover, header, footer and Document Governance 'Document' row were separately confirmed to read NCIE-013, not NCIE-012.

23. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python exec() call with an explicit encoding='utf-8' argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-013 document confirmed zero mojibake occurrences across all 680 paragraphs and 44 tables.

24. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 59 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone

and Document Governance begins on page 2 (the manual page break), with the running header and footer correctly reading NCIE-013 branding and the §0.1 title-reconciliation note visible from page 2 onward. No table clipping, overlapping text, or broken table was observed across the five consolidated matrices, the Chapter 33 Human Review Register tables, or the Chapter 34 traceability/handover tables.

25. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern HR13-<chapter>-<n>: TOTAL IDS FOUND = 34. UNIQUE IDS = 34. DUPLICATES = 0. OPEN = 33. RESOLVED = 1 (HR13-34-1, Resolved by Upstream Baseline). BLOCKING = 30. NON-BLOCKING = 4. CLASSIFICATION TOTALS: Institutional Confirmation Required = 22; Security/Sovereignty Confirmation Required = 5; Legal/Policy Confirmation Required = 3; Engineering/Infrastructure Discovery Required = 3; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 34 unique IDs found anywhere in the document also appears as a row in the Chapter 33 Complete Register table, which itself contains exactly 34 rows with no duplicate Review ID.

26. Three-Artifact Consistency Check

The NCIE-013 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Formal title 'NCIE-013 — Functional Modules Specification' (corrected per §1 above, SOURCE-VERIFIED against Master Controlled Register v2.0); Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-012 v1.0, plus the approved NCIE-013 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 34 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 34/34 explicit subsections present; Human Review Register TOTAL = 34, BLOCKING = 30, NON-BLOCKING = 4, classification totals as in §25 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; domain test-class status = 8 tests specified, none executed, passed or scenario-validated; functional-module invariants = 18 NCIE-013-specific plus cross-referenced upstream invariants; downstream handover = NCIE-015/016 each receive a defined interface without technical preemption; title-reconciliation result = SEMANTIC MATCH / DOCUMENT-CONTROL WORDING DIFFERS, corrected. No mismatch was found between the three artifacts.

27. Final Readiness Disposition

NCIE-013 — Functional Modules Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed rule catalogue, threshold, typology and ownership assignment deferred in Chapters 1-32 remains a proposal pending resolution of the Chapter 33 Human Review Register (including the acceptance-authority/ownership confirmation, HR13-1-1); every §32.1 test remains SPECIFIED, not executed or scenario-validated, pending NCIE-016. This document is NOT marked Approved.

28. Pre-Approval Title-Reconciliation Correction (Surgical, Post-Delivery)

This Verification Report was updated under a pre-approval surgical correction to v1.0 (not a v1.0→v1.1 amendment) after the originally delivered edition was found to state, incorrectly, that no Master Controlled Register was supplied for title reconciliation. The correction is documented in full in §1 above and in the accompanying Full-Expansion Change Register's Pre-Approval Surgical Correction Record; both the defect and its correction remain part of this document's traceable history rather than being erased. Scope of the correction was limited to source-availability correction, formal-title reconciliation, and document-control consistency; no chapter content, domain-module architecture, rule/KPI framework, or Human Review Register total was altered, added, or resolved by this correction.